Annex D (normative): Control sequences, using CSI

The (new) alphabets in subsection A.4 have a control code CSI, CONTROL SEQUENCE INTRODUCER. It comes from the standard ECMA-48, also ISO/IEC 6429 which is technically the same standard in ISO/IEC dress. Some CSI based control sequences are commonly implemented in terminal emulators, and some of the control sequences are for use in terminal emulators only. However, we are here (Annex D and E) referring to control sequences that are useful also outside of terminal emulators.

We will use the following general syntax for control sequences:

control-sequence_s ::= **CSI**[0-9:;=?]*[@A-Z_a-z]

This is slightly more limited compared to the ECMA-48 specification, but quite sufficient here. Indeed, we will only use the final characters "_" and "m". Note that we refer to characters, not byte codes.

Any error compared to this syntax (i.e. start with CSI, but then not fulfill the given syntax) results in REPLACEMENT CHARACTER for the CSI and as is for the rest of the sequence. Other syntax errors, specific to "_" or "m" also result in REPLACEMENT CHARACTER for the CSI and as is for the rest. So does, for SMS and CBS, using other terminating characters than "_" or "m". (Other systems may support more control sequences than those defined for SMS/CBS.)

D.1 Decimal character references

Unicode character references, in ECMA-48 style, for (Unicode) scalar values greater than U+009F. This mechanism is similar to HTML's decimal numeric character references. A decimal character reference as a control sequence, compatible with ECMA-48 (but not included in ECMA-48, since the last update of ECMA-48 was in 1976, long before Unicode (and ISO/IEC 10646) first versions):

CSI*n*_, where *n* is a Unicode scalar value in decimal form (using characters 0-9), without leading zeroes and only referring to a valid Unicode scalar value (i.e. less than 1114112 and not referring to a "surrogate") and is larger than 159 (decimal), but subject to the mappings in subsection 6.2.3. This form of character reference is similar to HTML's (decimal) character references: &#n;, using **CSI** instead of &# and _ instead of ;.

Note that hexadecimal cannot be used inside a control sequence due to the ECMA-48 syntax, where A-F (and a-f) are control sequence terminating characters.

This mechanism can be used to refer to a character that is not included in the used 7-bit encoding. It can be used for U+0127 (LATIN SMALL LETTER H WITH STROKE), ħ: **CSI**295**_**, used in Maltese, without needing to go over to UTF-16BE representation for the (sub)message. (Note that ħ is included in the SS3 table for alphabet 0x10.)

However, there should be only a small number of such character references in a message, since each take up several 7-bit code units. At some point it is more advantageous to use UTF16BE (or possibly another 7-bit alphabet, depending).

The interpretation of control sequences, like **CSI***n*_, must be done after the entire message is composed from all of the submessages it is composed from. If an implementation allows for partial display of a message during the collection of submessages, then only the complete control sequences are interpreted. Partial ones result in REPLACEMENT CHARACTER being displayed.

D.2 Text styling

Styling text using the mechanisms specified in ECMA-48 (ISO/IEC 6429) is popularly implemented in terminal emulators. But styling text is of course not limited to terminal emulators, nor is that particular mechanism as such limited to terminal emulators. However, styling by the mechanisms afforded by HTML/CSS cannot be used for SMS/CBS. Way too verbose, and not compatible with “plain text”. However, the mechanism afforded by ECMA-48 is compatible with “plain text” (which is why it at all can be used in terminal emulators), and thus compatible with SMS/CBS messages.

The text style (and text colour) are given as “style state changes” and can be combined. E.g. bold and underlined can be combined, italics and red can be combined. Changes in one text style (or text colour) aspect does not change any other text style/colour aspect.

Note that the space after “CSI” is for readability in the subsections below. It must not occur in the actual control sequence.

“Short codes” (3 7-bit code units each) for 7-bit alphabets are given in brackets.

Support for text styling/colouring is optional. This section obsoletes section 9.2.3.24.10.1.1 (Text Formatting) of 3GPP TS 23.040 (Technical realization of the Short Message Service (SMS)).

The text styling/colouring specified here is an “extended subset” of the styling/colouring controls from ISO/IEC 6429 (ECMA-48).

D.2.1 Bold

- **CSI 1m** Bold font variant. ”Medium” boldness.
  [0x1B1B00]
- **CSI 22m** Normal weight (not bold). (Default.)
  [0x1B1B01]

Note: there is no change in nominal colour in any way for these font weight changes, other than the overall optical change due to the change in weight.

D.2.2 Italic/oblique

- **CSI 3m** Italicized or oblique. Serifed fonts usually have a special variant for italics. Sans-serif fonts are often just slanted (oblique) relative to the upright variant. The slant should be around −7° to −10°.
  [0x1B1B02]
- **CSI 23m** Upright (a.k.a. roman). (Default.)
  [0x1B1B03]

D.2.3 Raised to superscript position

Proper superscript styling is not part of ECMA-48 5^{th} edition (from 1976), but it is a common typographical styling with meaning. Common enough to warrant inclusion here (for SMS messages). These superscripts are not really intended for math expressions, but may still be used for exponentiation of units, as can the characters ^{2} and ^{3}, included in all the new alphabets above.

- **CSI 56:1m** Raised to first superscript level and slightly smaller, about 80% current set font size. Used for ordinals (e.g. 5^{th}, 1^{er}), and other uses (e.g. M^{lle}, XX^{e}).
  [0x1B1B04]
- **CSI 56m** Not raised and use to the set font size. (Default.) This reset is auto-applied just before LF, SP and punctuation/symbol (except if a **CSI 56:1m** is just before the punctuation/symbol).
  [0x1B1B05]

The two “ordinal indicator” letters (included in the 0x10 alphabet), mainly for Iberian languages, should look just like the proper display of CSI 56:1m**a**CSI 56m and CSI 56:1m**o**CSI 56m. Likewise for ^{2} and ^{3}.

D.2.4 Underline

- **CSI 4m** Underlined. Single underline of “medium” weight.
  [0x1B1B06]

- **CSI 24m** Not underlined. (Default.)
  [0x1B1B07]

D.2.5 Strike-through

- **CSI 9m** Strike-through. Single strike-through of “medium” weight.
  [0x1B1B08]

- **CSI 29m** Not strike-through. (Default.)
  [0x1B1B09]

D.2.6 Font size (relative to default; changes do not accumulate)

For MEs that are capable of different sizes. If the ME is not capable of different font sizes, these are no-ops.

- **CSI 77:0?8m** Somewhat smaller than default, about 0.8 times default size.
  [0x1B1B0A]

- **CSI 77m** Default font size. (Default.) [**CSI 77:1m**]
  [0x1B1B0B]

- **CSI 77:1?2m** Somewhat larger than default, about 1.2 times default size.
  [0x1B1B0C]

- **CSI 77:1?4m** Larger than default, about 1.4 times default size.
  [0x1B1B0D]

D.2.7 Font type (actual font is implementation defined)

For MEs that are capable of different fonts. If the ME is not capable of different fonts, these are no-ops.

- **CSI 26:1m** Serifed proportional font.
  [0x1B1B0E]

- **CSI 26:2m** Sans-serif proportional font.
  [0x1B1B0F]

- **CSI 26m** Default proportional font. (Default if ME capable of proportional fonts.) [**CSI 26:0m**]
  [0x1B1B1F]

- **CSI 50m** Typewriter font (“fixed width”). (Default if ME only capable of “fixed width” fonts.)
  [0x1B1B2F]

D.3 Text colouring

Care should be taken (by the message author) so that the colours chosen results in a readable text, rather than a hidden or hard to read text. "Night mode" may 'photo-invert' the colours or may just change the default colours.

For MEs that are not capable of RGB colour, all "marker pen" colours (except fully transparent) are light grey (assuming a light background and black text), and text "foreground" colours are "b/w photo greyscaled"; in case greyscale is not possible, all text colouring are no-ops.

"Short codes" (3 7-bit code units each) for 7-bit alphabets are given in brackets.

D.3.1 Text "marker pen" colouring

For truly black and white (no greyscale) displays, the colour changes have no effect (not even black and white). For greyscale, all the colours here (even yellow, black, white) may be displayed as light grey without transparency.

Text background colour or better referred to as text highlight colour or "marker pen" colour. The default 'colour' for this is fully transparent (so one gets the background colour that is "behind" the text block). This colour "plane" is "below" the text "foreground" colour, but "above" the text block background.

Colour is given as RGB (Red, Green, Blue) additive colour component values; 0 is none, 255 is max. Transparency for this highlight ("marker pen") colour is implementation defined, like 20 % transparent, but could be opaque. The colours in the list are too "harsh" as is as marker pen colouring, but with transparency they are more "marker pen"-like.

CSI 40m : RGB: **0:0:0**	Pure black ■	[0x1B1B10]
CSI 41m : RGB: **205:0:0**	Dull red ■	[0x1B1B11]
CSI 42m : RGB: **0:205:0**	Dull green ■	[0x1B1B12]
CSI 43m : RGB: **255:215:0**	Dull yellow ■	[0x1B1B13]
CSI 44m : RGB: **0:0:205**	Dull blue ■	[0x1B1B14]
CSI 45m : RGB: **205:0:205**	Dull magenta ■	[0x1B1B15]
CSI 46m : RGB: **0:205:205**	Dull cyan ■	[0x1B1B16]
CSI 47m : RGB: **255:255:255**	Pure white	[0x1B1B17]
CSI 49m : Fully transparent (default); transparency **255**.		[0x1B1B18]
CSI 100m : RGB: **105:105:105**	Dark grey ■	[0x1B1B19]
CSI 101m : RGB: **255:0:0**	Clear red ■	---
CSI 102m : RGB: **0:255:0**	Clear green ■	---
CSI 103m : RGB: **255:255:0** (not recommended)	Clear yellow ■	---
CSI 104m : RGB: **0:0:255**	Clear blue ■	[0x1B1B1A]
CSI 105m : RGB: **255:0:255**	Clear magenta ■	---
CSI 106m : RGB: **0:255:255**	Clear cyan ■	[0x1B1B1B]
CSI 107m : RGB: **220:220:220**	Light grey ■	[0x1B1B1C]
CSI 108m : RGB: **169:169:169**	Medium grey ■	[0x1B1B1D]
CSI 109m : RGB: **255:140:0**	Orange ■	[0x1B1B1E]

The space after "CSI" is for readability here. It must not occur in the actual control sequence. The 108 and 109 are additions compared to existing terminal emulators.

D.3.2 Text colour ("foreground" colour)

Colour for the text itself. For truly black and white (no greyscale) displays, the colour changes have no effect (not even black and white). For greyscale, all the colours here (even yellow, black, white) may be displayed as dark grey. The colour of underlines (if any) follows the colour of the text.

These colours should not have any transparency (though that is implementation defined). Note that this colour is "above" the "marker pen"-like colouring. These colours are not transparent. The colour RGB values is a compromise between the existing terminal emulators (with a preference for xterm) colour RGB values. The 98 and 99 are additions compared to existing terminal emulators.

CSI 30m : RGB: **0:0:0** (common default)	Pure black ■	[0x1B1B20]
CSI 31m : RGB: **205:0:0**	Dull red ■	[0x1B1B21]
CSI 32m : RGB: **0:205:0**	Dull green ■	[0x1B1B22]
CSI 33m : RGB: **255:215:0**	Dull yellow ■	[0x1B1B23]
CSI 34m : RGB: **0:0:205**	Dull blue ■	[0x1B1B24]
CSI 35m : RGB: **205:0:205**	Dull magenta ■	[0x1B1B25]
CSI 36m : RGB: **0:205:205**	Dull cyan ■	[0x1B1B26]
CSI 37m : RGB: **255:255:255**	Pure white	[0x1B1B27]
CSI 39m : Default text colour (implementation defined)		[0x1B1B28]
CSI 90m : RGB: **105:105:105**	Dark grey ■	[0x1B1B29]
CSI 91m : RGB: **255:0:0**	Clear red ■	---
CSI 92m : RGB: **0:255:0**	Clear green ■	---
CSI 93m : RGB: **255:255:0** (not recommended)	Clear yellow ■	---
CSI 94m : RGB: **0:0:255**	Clear blue ■	[0x1B1B2A]
CSI 95m : RGB: **255:0:255**	Clear magenta ■	---
CSI 96m : RGB: **0:255:255**	Clear cyan ■	[0x1B1B2B]
CSI 97m : RGB: **220:220:220**	Light grey ■	[0x1B1B2C]
CSI 98m : RGB: **169:169:169**	Medium grey ■	[0x1B1B2D]
CSI 99m : RGB: **255:140:0**	Orange ■	[0x1B1B2E]

The space after “CSI” is for readability here. It must not occur in the actual control sequence.

Usage advice: the text marking and text colour settings should have good contrast, i.e. not try to hide the text or make it hard to read; at least when assuming reasonable default colours (light background (incl. marking), dark text).

D.3.3 Style control abbreviations

A sequence of style setting can be put together in a single CSI-sequence, with semicolon as separator. E.g. bold (CSI1m) clear red (CSI91m), can be specified as **CSI**1;91m. For 7-bit alphabets one can often use the 3 7-bit code unit abbreviation codes listed above.

An empty sequence, CSIm, is not a no-op, instead it resets all styles to default style (upright, normal weight, not underlined, not superscript, default colours). For 7-bit alphabets, LF is (non-recursively) interpreted as **CSI**m**LF**.

These abbreviations, except for the 7-bit alphabet ones, are specified in ECMA-48 (ISO/IEC 6429) and have been commonly implemented in terminal emulators.